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Combinatorics
Practice Assignment • 40 min

Due Jul 10, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. How many numbers must be selected from the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ to guarantee that at least one pair of these numbers adds up to 13? 1 / 1 point

- Four
- Six
- 7
- Two

Nice work

It is correct. There are six pairs and their sum is 13, (1,12), (2,11),(3,10),(4,9),(5,8) and (6,7). So, we have six pigeonholes. Now, if we choose seven numbers (pigeons), according to the Pigeonhole Principle there are at least two numbers among these numbers that adds up to 13.

2. How many binary strings of length 7 have more 0s than 1s? 1 / 1 point

- 63
- 56
- 64
- 259

Nice work

It is correct. We need to find the strings that have four, five, six and seven zeros. The answer is the sum of them. The number of strings that have four zeros are C(7,4), five zeros C(7,5), six zeros C(7,6) and seven zeros C(7,7). The answer is 35+21+7+1=64.

3. A coin is flipped seven times. How many possible outcomes contains at least three tails? 1 / 1 point

- 99
- 35
- 64
- 98

Nice work

It is correct. We should have at least three tails, which means we should have three, four, five, six or seven tails. The number of possible outcomes having three tails is C(7,3), four tails C(7,4), five tails C(7,5), six tails C(7,6) and seven tails C(7,7). The answer is the sum of these numbers: 35+35+21+7+1=99.

4. How many hands of four can be dealt from a deck of 10 cards? 1 / 1 point

- 5,040
- 10,000
- None of these answers.
- 210

Nice work

It is correct. It is C(10,4), which is 210.

5. How many non-repetitive sequences of length 3 can be made from 10 digits? 1 / 1 point

Note: 002 is not acceptable because 0 occurs twice.

- 720
- 120
- 72
- 10,00

Nice work

It is correct. The answer is P(10,3) which is 720.

6. How many non-repetitive sequences of length 6 can be made from 10 digits that contains '42'? 1 / 1 point

Note: 002142 is not acceptable because 0 occurs twice and 456782 is not acceptable because it does not contain 42.

- 1,680
- 8,400
- 151,200
- 20,160

Nice work

It is correct. Considering 42 as a block, it can be in position 1, 2, 3, 4 or 5. So, there are five cases. In each case, there are P(8,4) as there are 8 remaining digits and 4 remaining spaces in the sequence. The overall sequences are 5*P(8,4) which is 5*(5*6*7*8)=8400.

7. How many non-repetitive sequences of length at most 5 can be made from 10 digits that contains '42'? 1 / 1 point

Note: 0021 is not acceptable because 0 occurs twice.

- 1,344
- 1,529
- 36,100
- 401

Nice work

'At most 5' means sequences with length 1, 2, 3, 4 and 5. Since the sequence must contain 42, sequences with length 1 cannot be accepted. The number of sequences with length 2 is P(8,0), where '8' is the number of remaining digits and '0' is the number of remaining spaces in the sequence. The number of sequences with length 3 is P(8,1)*2, 4 is P(8,2)*3 and 5 is P(8,3)*4. The answer is the sum of these numbers, which is 1+16+168+1344=1529.

8. How many positive numbers less than 150 are divisible by either 2 or 3? 1 / 1 point

- 99
- 100
- 125
- 123

Nice work

It is correct. There are 149/2=74 positive numbers divisible by 2 and 149/3=49 divisible by 3, so in total 74+49=123, but there are 149/6=24 numbers that are divisible by both 2 and 3 so we have counted them twice. Thus, the final answer is 123-24=99.